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1  theory HipSort
2    imports Main "HOL-Library.Multiset"
3  begin
4
5  definition swap :: "'a list  $\Rightarrow$  nat  $\Rightarrow$  nat  $\Rightarrow$  'a list" where
6    "swap l i j = l[i := l ! j, j := l ! i]"
7
8  fun roditelj :: "nat  $\Rightarrow$  nat" where
9    "roditelj i = (i - 1) div 2"
10
11 fun levo :: "nat  $\Rightarrow$  nat" where
12   "levo i = 2*i + 1"
13
14 fun desno :: "nat  $\Rightarrow$  nat" where
15   "desno i = 2*i + 2"
16
17 lemma mset_swap [simp]:
18 assumes "i < length l"
19 and "j < length l"
20 shows "mset (swap l i j) = mset l"
21   using assms
22   unfolding swap_def
23   by (metis mset_swap)
24
25 (*grupa ubaci*)
26 fun ubaci :: "int list  $\Rightarrow$  nat  $\Rightarrow$  int list" where
27   "ubaci l i =
28     (if i = 0 then
29       l
30     else
31       (if l ! (roditelj i)  $\geq$  l ! i then
32         l
33       else
34         ubaci (swap l i (roditelj i)) (roditelj i)))"
35
36 (*grupa ubaci*)
37 function ubaciSve :: "int list  $\Rightarrow$  nat  $\Rightarrow$  int list" where
38   "ubaciSve l i = (if i  $\geq$  length l then l else ubaciSve (ubaci l i) (i+1))"
39   by pat_completeness auto
40 termination
41   sorry
42
43 fun najveci3 :: "int list  $\Rightarrow$  nat  $\Rightarrow$  nat  $\Rightarrow$  nat" where
44   "najveci3 l i m =
45     (if desno i < m then
46       (if l ! i  $\geq$  l ! levo i  $\wedge$  l ! i  $\geq$  l ! desno i then
47         i
48       else if l ! levo i  $\geq$  l ! desno i then
49         levo i
50       else
51         desno i
52     )
53   else if levo i < m then
54     (if (l ! i)  $\geq$  (l ! levo i) then
55       i
56     else
57       levo i
58   )
59   else

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60     i
61   )"
62
63 (*grupa izbaci*)
64 function izbaci :: "int list  $\Rightarrow$  nat  $\Rightarrow$  nat  $\Rightarrow$  int list" where
65 "izbaci l i m = (let najveci = najveci3 l i m in
66   (if i = najveci then l
67     else izbaci (swap l i najveci) najveci m))"
68 by pat_completeness auto
69 termination
70 sorry
71
72 (*grupa izbaci*)
73 fun izbaciSve :: "int list  $\Rightarrow$  nat  $\Rightarrow$  int list" where
74 "izbaciSve l i = (if i = 0 then l else izbaciSve (izbaci (swap l 0 (i-1)) 0 (i-1)) (i-1))"
75
76 fun HipSort :: "int list  $\Rightarrow$  int list" where
77 "HipSort l = izbaciSve (ubaciSve l 0) (length l)"
78
79 (*grupa ubaci*)
80 fun JesteHip1 :: "int list  $\Rightarrow$  nat  $\Rightarrow$  bool" where
81 "JesteHip1 l m = ( $\forall i \in \{1..<m\}$ . l!roditelj i  $\geq$  l!i)"
82
83 (*grupa izbaci*)
84 fun JesteHip2 :: "int list  $\Rightarrow$  nat  $\Rightarrow$  bool" where
85 "JesteHip2 l m = ( $\forall i \in \{0..<m\}$ . i = najveci3 l i m)"
86
87 lemma l2tol:
88   assumes "0 < i"
89   and "i < m"
90   and "najveci3 l (roditelj i) m = roditelj i"
91   shows "l ! i  $\leq$  l ! roditelj i"
92 proof (cases "2 dvd i")
93   case True
94   have "roditelj i < m"
95     using assms
96     by (metis One_nat_def Suc_leI True add_Suc_right diff_le_self div_less_dividend
97       lessI nat_arith.rule0
98       nat_dvd_not_less nat_less_le one_add_one order_less_le_trans roditelj.simps
99       zero_less_diff)
100   then have "roditelj i = najveci3 l (roditelj i) m"
101     using assms
102     by auto
103   moreover
104     have "i = desno (roditelj i)"
105     using assms True
106     by (metis One_nat_def Suc_pred add.commute add_gr_0 desno.elims div2_Suc_Suc dvdE
107       even_Suc_div_two lessI
108       mult_Suc_right nat_less_le nonzero_mult_div_cancel_left one_add_one
109       roditelj.simps)
110   ultimately show ?thesis
111   by (smt (verit, ccfv_SIG) assms najveci3.simps)
112 next
113 case False
114 have "roditelj i < m"
115   using assms
116   by (metis (no_types, lifting) One_nat_def Suc_le_eq Suc_pred add_Suc_right div_less
117     div_less_dividend)

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113      linorder_not_less nat_arith.rule0 nat_less_le one_add_one order_less_le_trans
      roditelj.simps)
114 then have "roditelj i = najveci3 l (roditelj i) m"
115   using assms
116   by metis
117 moreover
118 have "i = levo (roditelj i)"
119   using assms False
120   by auto
121 ultimately show ?thesis
122   by (smt (verit, ccfv_SIG) assms najveci3.simps)
123 qed
124
125 lemma najveci3slucaj1:
126   assumes "desno i < m"
127   shows "(najveci3 l i m = i)  $\longleftrightarrow$  (l!i  $\geq$  l!levo i  $\wedge$  l!i  $\geq$  l!desno i)"
128   using assms
129   by auto
130
131 lemma najveci3slucaj2:
132   assumes "desno i = m"
133   shows "(najveci3 l i m = i)  $\longleftrightarrow$  (l!i  $\geq$  l!levo i)"
134   using assms
135   by auto
136
137 lemma najveci3slucaj3:
138   assumes "desno i > m"
139   shows "najveci3 l i m = i"
140   using assms
141   by auto
142
143 lemma l1to2:
144   assumes "(desno i < m  $\wedge$  l!i  $\geq$  l!levo i  $\wedge$  l!i  $\geq$  l!desno i)  $\vee$  (desno i = m  $\wedge$  l!i  $\geq$ 
      l!levo i)  $\vee$  (desno i > m)"
145   shows "najveci3 l i m = i"
146   using assms
147   by auto
148
149 lemma JesteHipEquivDef: "JesteHip1 l m = JesteHip2 l m"
150 proof
151   assume "JesteHip1 l m"
152   then have *: " $\forall i. (0 < i \wedge i < m) \longrightarrow l!(roditelj\ i) \geq l!i$ "
153     by auto
154   have " $\forall i. i < m \longrightarrow i = najveci3\ l\ i\ m$ "
155   proof
156     fix i::nat
157
158     show "i < m  $\longrightarrow$  i = najveci3 l i m "
159       using * l1to2
160       by auto
161   qed
162   then show "JesteHip2 l m"
163     by auto
164 next
165   assume "JesteHip2 l m"
166   then have *: " $\forall i. i < m \longrightarrow i = najveci3\ l\ i\ m$ "
167     by auto
168   have " $\forall i. (0 < i \wedge i < m) \longrightarrow l!(roditelj\ i) \geq l!i$ "
169   proof

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170   fix i::nat
171
172   show " $0 < i \wedge i < m \longrightarrow l ! i \leq l ! \text{rodite}l j i$ "
173     using l2tol *
174     by auto
175   qed
176   then show "JesteHip1 l m"
177     by auto
178 qed
179
180 fun najveci3rodite l j i m :: "int list  $\Rightarrow$  nat  $\Rightarrow$  nat  $\Rightarrow$  nat" where
181 "najveci3rodite l j i m =
182   (if desno i < m then
183     (if l!rodite l j i  $\geq$  l!levo i  $\wedge$  l!rodite l j i  $\geq$  l!desno i then
184       rodite l j i
185     else if l!levo i  $\geq$  l!desno i then
186       leve i
187     else
188       desno i
189   )
190   else if leve i < m then
191     (if (l!rodite l j i)  $\geq$  (l!levo i) then
192       rodite l j i
193     else
194       leve i
195   )
196   else
197     rodite l j i
198 )"
199
200 (*grupa ubaci*)
201 fun SkoroHip1 :: "int list  $\Rightarrow$  nat  $\Rightarrow$  nat  $\Rightarrow$  bool" where
202 "SkoroHip1 l m q = (( $\forall i \in \{1..m\} - \{q\}. l!rodite l j i \geq l!i$ )  $\wedge$  najveci3rodite l j q m =
   rodite l j q)"
203
204 (*grupa izbaci*)
205 fun SkoroHip2 :: "int list  $\Rightarrow$  nat  $\Rightarrow$  nat  $\Rightarrow$  bool" where
206 "SkoroHip2 l m q = ( $\forall i \in \{0..m\} - \{q\}. i = \text{najveci3 } l i m$ )"
207
208 lemma VezaSkoroJeste2: " $2 * q + 1 > m \wedge \text{SkoroHip2 } l m q \longrightarrow \text{JesteHip2 } l m$ "
209 proof
210   assume *: " $m < 2 * q + 1 \wedge \text{SkoroHip2 } l m q$ "
211   from * have " $\forall i \in \{0..m\} - \{q\}. i = \text{najveci3 } l i m$ "
212     by auto
213   moreover
214   from * have " $q = \text{najveci3 } l q m$ "
215     by auto
216   ultimately show "JesteHip2 l m"
217     by (metis JesteHip2.elims(3) insert_iff insert_Diff_single)
218 qed
219
220 fun JesteSortiran :: "int list  $\Rightarrow$  nat  $\Rightarrow$  bool" where
221 "JesteSortiran l m = sorted (drop m l)"
222
223 value "JesteHip [9, 3, 8, 2, 3, 8, 8, 1, 2, 3, 1, 8] 10"
224 value "najveci3 [9, 3, 7, 8, 2, 3, 5, 3, 8, 8] 1 4"
225
226 (*
227 lemma swap_lemma1:

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228 assumes "i < length l"
229 and "j < length l"
230 shows " $\forall k \in \{0..<\text{length } l\} - \{i, j\}. l!k = (\text{swap } l \ i \ j)!k$ "
231 unfolding swap_def
232 by auto
233
234 lemma swap_lemma2:
235   assumes "i < length l"
236   and "j < length l"
237   shows "(swap l i j)!i = l!j"
238   and "(swap l i j)!j = l!i"
239   unfolding swap_def
240   using assms
241   sledgehammer
242   by (metis list_update_id nth_list_update_eq nth_list_update_neq, simp)
243*)
244
245
246 lemma ubaci_SkoroHip1:
247   assumes "SkoroHip1 l m q"
248   and "q  $\neq$  0"
249   and "q < m"
250   and "l ! roditelej q < l ! q"
251   and "nl = swap l q (roditelej q)"
252   and "m  $\leq$  length l"
253   shows "SkoroHip1 nl m (roditelej q)"
254 proof -
255   from assms(1) have tv1: " $\forall i \in \{1..<m\} - \{q\}. l!roditelej \ i \geq l!i$ "
256   by auto
257
258   from assms(2, 3, 5, 6) have swap1: "l!roditelej q = nl!q"
259   unfolding swap_def
260   by auto
261   from assms(2, 3, 5, 6) have swap2: "nl!roditelej q = l!q"
262   unfolding swap_def
263   by auto
264   from assms(2, 3, 5, 6) have swap3: " $\forall x. x < m \wedge x \neq q \wedge x \neq roditelej \ q \longrightarrow nl!x = l!x$ "
265   unfolding swap_def
266   by auto
267
268   from assms(1) have tv2: "najveci3roditelej l q m = roditelej q"
269   by auto
270   have tv3: "najveci3 nl q m = q"
271   proof (cases "desno q < m")
272     case True
273     with tv2 have "l!roditelej q  $\geq$  l!levo q  $\wedge$  l!roditelej q  $\geq$  l!desno q"
274     by (smt (verit, best) najveci3roditelej.elims)
275     with swap1 swap3 True have "nl!q  $\geq$  nl!levo q  $\wedge$  nl!q  $\geq$  nl!desno q"
276     by auto
277     then show ?thesis
278     by auto
279   next
280     case False
281     then show ?thesis
282   proof (cases "levo q < m")
283     case True
284     with tv2 have "l!roditelej q  $\geq$  l!levo q"
285     by (smt (verit, best) najveci3roditelej.elims)

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286     with swap1 swap3 True have "nl!q ≥ nl!levo q"
287     by auto
288     with False show ?thesis
289     by auto
290   next
291     case False
292     then show ?thesis
293     by auto
294   qed
295 qed
296
297
298
299 have "najveci3roditelj nl (roditelj q) m = roditelj (roditelj q)"
300 proof (cases "2 dvd q")
301   case True
302   have "roditelj q < m"
303   using assms(3)
304   by auto
305   then have "roditelj q = najveci3 l (roditelj q) m"
306   using tv1
307 next
308   case False
309   then show ?thesis sorry
310 qed
311
312
313
314
315 have "najveci3 l q m = q"
316 proof (cases "desno q < m")
317   case True
318
319   have hL: "l!q ≥ l!(levo q)"
320   proof -
321     from True assms(3) tv1 have "l!roditelj (levo q) ≥ l!(levo q)"
322     by fastforce
323     then show ?thesis
324     by auto
325   qed
326
327   have hR: "l!q ≥ l!(desno q)"
328   proof -
329     from True assms(3) tv1 have "l!roditelj (desno q) ≥ l!(desno q)"
330     by fastforce
331     then show ?thesis
332     by auto
333   qed
334
335   from hL hR show ?thesis
336   by auto
337 next
338   case False
339   show ?thesis
340 proof (cases "levo q < m")
341   case True
342
343   have hL: "l!q ≥ l!(levo q)"
344   proof -

```

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345     from True assms(3) tv1 have "l!roditelj (levo q) ≥ l!(levo q)"
346     by fastforce
347     then show ?thesis
348     by auto
349   qed
350   from hL False show ?thesis
351   by auto
352 next
353   case False
354   then show ?thesis
355   by auto
356 qed
357 qed
358
359 with assms(4, 5) swap1 swap2 have "najveci3roditelj nl q m = roditelj q"
360   unfolding swap_def
361   by (smt (verit, ccfv_SIG) najveci3.simps najveci3roditelj.elims nth_list_update_neq)
362
363
364 qed
365
366 lemma ubaci_JesteHip1:
367   assumes "SkoroHip1 l m q"
368   and "q < m"
369   and "q = 0 ∧ l ! roditelj q ≥ l ! q"
370   shows "JesteHip1 l m"
371 proof
372
373   qed
374
375 lemma ubaciSve_inv1:
376   assumes "i ≤ length l" (??)
377   and "JesteHip l i"
378   shows "JesteHip (ubaci l i) (i+1)"
379   sorry
380
381 lemma ubaciSve_inv2:
382   assumes "i ≤ length l" (??)
383   shows "mset l = mset (ubaci l i)"
384   unfolding swap_def
385   sorry
386
387
388 lemma izbaciSve_inv1:
389   assumes "0 < i ∧ i ≤ length l" (??)
390   and "JesteHip l i"
391   and "JesteSortiran l i"
392   and "l ! 0 ≤ l ! i"
393   and "nl = (swap l 0 (i-1))"
394
395   shows "JesteHip (izbaci nl 0 (i-1)) (i-1)"
396   and "JesteSortiran (izbaci nl 0 (i-1)) (i-1)"
397   and "nl ! 0 ≤ nl ! (i-1)"
398   sorry
399
400 lemma izbaciSve_inv2:
401   assumes "0 < i ∧ i ≤ length l" (??)
402   shows "mset l = mset (izbaci (swap l 0 (i-1)) 0 (i-1))"
403   unfolding swap_def

```

```
404 sorry  
405  
406  
407end
```